

# Software Management System

## System Design Document

### 1. Overview

The Software Management System is a branch-based ticket approval and project delivery workflow. Each branch has its own Branch Manager(s) and Executive Officer(s). A Branch Manager raises a project proposal as a ticket, which is reviewed first by that branch's Executive Officer and then by the IT Director. Once the IT Director authorizes it, the ticket becomes a project, which the IT Main Developer resources by assigning tasks to Developers. An Admin role manages branches, users, and developers outside this approval chain.

### 2. Actors and Roles

| Role              | Responsibilities                                                                                                                                                                                                                           |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Branch Manager    | Creates a ticket (project name + requirements) for their branch. Can edit the ticket while it is in Draft. Once sent, the ticket becomes view-only until it is either rejected back to them or completed.                                  |
| Executive Officer | Belongs to a branch (a branch can have more than one). Views, edits, and reviews tickets sent by that branch's Branch Manager(s). Approves or rejects, recording a remark that serves as the official decision minute to reject proposals. |
| IT Director       | Reviews tickets already approved by the Executive Officer. Approves (which authorizes the project) or rejects with a remark.                                                                                                               |
| IT Main Developer | Receives authorized projects. Assigns developers, creates/oversees the project, and tracks overall progress. Does not perform tasks directly.                                                                                              |
| Developer         | Views tasks assigned by the IT Main Developer and updates the status of only their own assigned tasks.                                                                                                                                     |
| Admin             | Manages branches, user accounts, and developers. Has visibility into the status of all tickets across all branches.                                                                                                                        |

#### 2.1 Branch Structure

Branches are set up first, independently of the ticket workflow. Each branch can have multiple Branch Managers and multiple Executive Officers; a ticket raised by any Branch Manager in a branch is reviewed by an Executive Officer belonging to that same branch.

### 3. End-to-End Workflow

- A Branch Manager creates a ticket with a project name and requirements. While in Draft, it can be freely edited.
- The Branch Manager sends the ticket to the Executive Officer of their branch. Its status moves to Pending Executive Review, and it becomes read-only for the Branch Manager from this point.
- The Executive Officer views, and if needed edits, the ticket, then approves or rejects it, recording a remark that serves as the official decision minute.
- If rejected by the Executive Officer, the ticket returns to the Branch Manager who sent it, together with the remark.
- If approved, the ticket moves to Pending Director Review and is forwarded to the IT Director.
- The IT Director reviews the ticket and approves or rejects it, again with a recorded remark.
- If rejected by the IT Director, the ticket is returned to the Executive Officer of that branch, who forwards it back to the Branch Manager.

- If approved by the IT Director, the ticket is authorized and becomes a Project, which is handed directly to the IT Main Developer.
- The IT Main Developer assigns developers to the project, tracks its progress, and creates/updates the tasks that make it up.
- Each Developer views their assigned task(s) and updates the status of only their own tasks (e.g. In Progress, Completed).
- The IT Main Developer tracks overall progress from the aggregated status of all tasks and marks the project complete once all tasks are done.
- Independently of this chain, the admin manages branch records and user accounts (including assigning roles and branches to new users) and can view the status of any ticket across the system.

## 4. Data Model (ER Design)

Eight entities make up the schema, matching the ER diagram. Primary keys are marked PK; foreign keys are marked FK. Fields not present on the original diagram, added to support login and auditability, are marked “Added”.

### 4.1 Branch

| Field       | Type         | Notes                 |
|-------------|--------------|-----------------------|
| BID         | INT, PK      | Auto-increment        |
| Branch_Name | VARCHAR(100) | e.g. “Finance Branch” |

### 4.2 User\_Types

| Field     | Type        | Notes                                                                               |
|-----------|-------------|-------------------------------------------------------------------------------------|
| Type_ID   | INT, PK     | Auto-increment                                                                      |
| User_Type | VARCHAR(50) | Branch Manager, Executive Officer, IT Director, IT Main Developer, Developer, Admin |

### 4.3 Users

| Field    | Type                       | Notes                                                        |
|----------|----------------------------|--------------------------------------------------------------|
| User_ID  | INT, PK                    | Auto-increment                                               |
| Type_ID  | INT, FK → User_Types       | Determines the user's role                                   |
| BID      | INT, FK → Branch, nullable | Null for IT-department roles not tied to one branch          |
| Email    | VARCHAR(150)               | Unique                                                       |
| Username | VARCHAR(100)               |                                                              |
| Password | VARCHAR(255)               | Added — hashed, never stored in plain text; needed for login |

### 4.4 Tickets

| Field     | Type             | Notes                             |
|-----------|------------------|-----------------------------------|
| Ticket_ID | INT, PK          | Auto-increment                    |
| BID       | INT, FK → Branch | Branch the ticket originates from |

| Field        | Type               | Notes                                                 |
|--------------|--------------------|-------------------------------------------------------|
| User_ID      | INT, FK → Users    | Branch Manager who created it                         |
| Project_Name | VARCHAR(150)       |                                                       |
| Requirements | TEXT               |                                                       |
| Status       | VARCHAR(30)        | See Section 5 for the full lifecycle                  |
| Created_At   | DATETIME           | Added — needed to track when the ticket was raised    |
| Sent_At      | DATETIME, nullable | Added — set when sent; editing locks once this is set |

#### 4.5 TicketDocument

| Field       | Type              | Notes                               |
|-------------|-------------------|-------------------------------------|
| Document_ID | INT, PK           | Auto-increment                      |
| Ticket_ID   | INT, FK → Tickets |                                     |
| File_Name   | VARCHAR(150)      |                                     |
| File_Path   | VARCHAR(255)      | Storage location                    |
| Uploaded_At | DATETIME          | Must be before the ticket's Sent_At |

#### 4.6 Ticket Decision / Approvals

| Field       | Type              | Notes                                                               |
|-------------|-------------------|---------------------------------------------------------------------|
| Decision_ID | INT, PK           | Auto-increment                                                      |
| Ticket_ID   | INT, FK → Tickets |                                                                     |
| User_ID     | INT, FK → Users   | The reviewer: Executive Officer or IT Director                      |
| Decision    | VARCHAR(20)       | Approved / Rejected                                                 |
| Decision_as | VARCHAR(50)       | Role the reviewer acted in (Executive Officer / IT Director)        |
| Remark      | TEXT              | Doubles as the official decision minute                             |
| Decision_At | DATETIME          | Added — replaces the unclear “User-G” field on the original diagram |

#### 4.7 Approved\_Projects

| Field        | Type                      | Notes                                    |
|--------------|---------------------------|------------------------------------------|
| Project_ID   | INT, PK                   | Auto-increment                           |
| Ticket_ID    | INT, FK → Tickets, unique | One-to-one; created on Director approval |
| Project_Name | VARCHAR(150)              | Copied from the ticket at creation       |
| Status       | VARCHAR(30)               | Not Started / In Progress / Completed    |

| Field      | Type     | Notes                                                   |
|------------|----------|---------------------------------------------------------|
| Created_At | DATETIME | Added — needed to track when the project was authorized |

## 4.8 Tasks

| Field       | Type              | Notes                                                                                         |
|-------------|-------------------|-----------------------------------------------------------------------------------------------|
| Task_ID     | INT, PK           | Auto-increment                                                                                |
| Ticket_ID   | INT, FK → Tickets | Links the task back to the project's originating ticket                                       |
| Assigned_To | INT, FK → Users   | The Developer assigned to this task (was MID → Project_Members; now points straight at Users) |
| Task_Title  | VARCHAR(150)      | Added — needed so a Developer knows what the task is                                          |
| Description | TEXT              | Added — task details                                                                          |
| Status      | VARCHAR(30)       | Not Started / In Progress / Completed                                                         |

## 4.9 Relationship Summary

| From              | Cardinality | To                          |
|-------------------|-------------|-----------------------------|
| Branch            | 1 → N       | Users                       |
| User_Types        | 1 → N       | Users                       |
| Branch            | 1 → N       | Tickets                     |
| Users (creator)   | 1 → N       | Tickets                     |
| Tickets           | 1 → N       | TicketDocument              |
| Tickets           | 1 → N       | Ticket Decision / Approvals |
| Users (reviewer)  | 1 → N       | Ticket Decision / Approvals |
| Tickets           | 1 → 1       | Approved_Projects           |
| Tickets           | 1 → N       | Tasks                       |
| Users (developer) | 1 → N       | Tasks                       |

## 5. Status Lifecycles

### 5.1 Ticket

| Status                   | Meaning                                                      |
|--------------------------|--------------------------------------------------------------|
| Draft                    | Being created/edited by the Branch Manager; not yet sent     |
| Pending Executive Review | Sent; awaiting the branch's Executive Officer decision       |
| Rejected by Executive    | Returned to the Branch Manager with a remark; editable again |
| Pending Director Review  | Approved by the Executive Officer; awaiting the IT Director  |

| Status                    | Meaning                                                                                                     |
|---------------------------|-------------------------------------------------------------------------------------------------------------|
| Rejected by Director      | Returned to the Executive Officer, who forwards it back to the Branch Manager with a remark; editable again |
| Approved / In Development | Authorized; a Project has been created                                                                      |
| Completed                 | All tasks under the linked project are complete                                                             |

## 5.2 Project

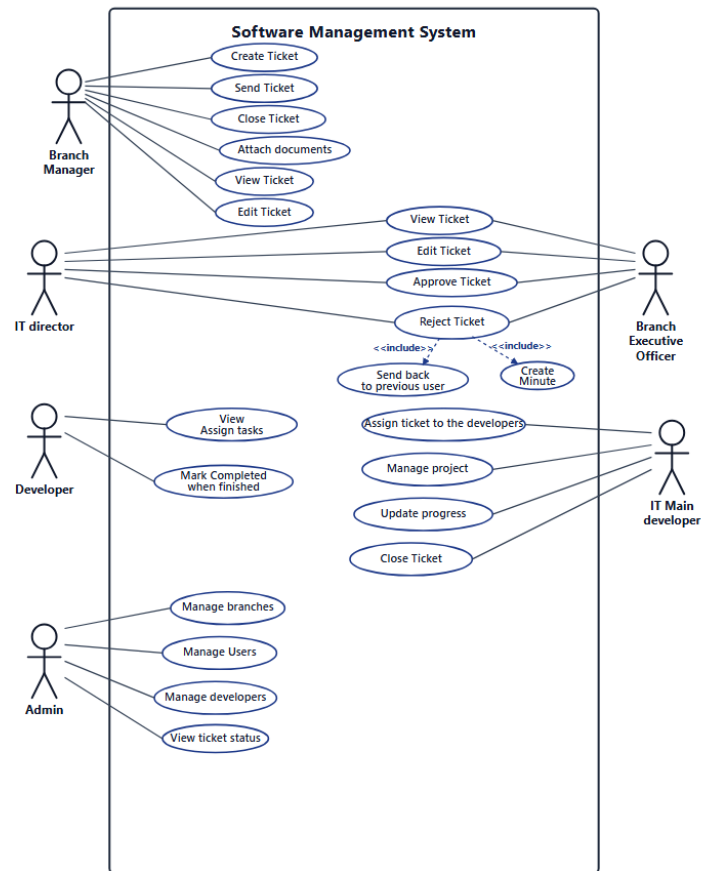
| Status      | Meaning                                |
|-------------|----------------------------------------|
| Not Started | Created, no tasks assigned yet         |
| In Progress | At least one task assigned or underway |
| Completed   | All tasks marked complete              |

## 5.3 Task

| Status      | Meaning                             |
|-------------|-------------------------------------|
| Not Started | Assigned, work not yet begun        |
| In Progress | Developer is actively working on it |
| Completed   | Developer has marked it done        |

## 6. Use Case Diagram

The diagram below shows each actor and the use cases they interact with, as sketched for this system.



## 7. Use Case Summary

| Actor             | Use Cases                                                                                         |
|-------------------|---------------------------------------------------------------------------------------------------|
| Branch Manager    | Create Ticket, Attach Documents, Edit Ticket (Draft only), Send Ticket, View Ticket, Close ticket |
| Executive Officer | View Ticket, Edit Ticket, Approve Ticket, Reject Ticket (includes Create Minute)                  |
| IT Director       | Review Approved Ticket, Approve Ticket (Authorize Project), Reject Ticket                         |
| IT Main Developer | Assign Developers, Create/Manage Project, Update Progress, Close Ticket                           |
| Developer         | View Assigned Tasks, Update Task Status, Mark Task Completed                                      |
| Admin             | Manage Branches, Manage Users, Manage Developers, View All Ticket Statuses                        |